

ASSIGNMENT - 2

FRACTIONS IN ANCIENT INDIAN MATHEMATICS

* FRACTIONS :-

The Sanskrit term for a fraction was *bhinna* which means "broken". The other terms for a fraction were *bhāga* and *aṁśa* meaning "part" or "portion". The term *kalā* which in Vedic times represented one-sixteenth was later used for a fraction too. *Caraka*, a commentator of *Līlāvati*, called a numerator *bhāga*, *aṁśa*, *vibhāga* or *laga* and the terms *hara*, *hāra* and *chheda* he used for a denominator.

In *Sūtra-sūtra*, unit fractions were named by a number with the term *bhaga* or *aṁśa*, thus *pañca-bhāga* (five parts) was the name of $1/5$. Sometimes fractions were denoted by an ordinal number with the term *bhaga* or *aṁśa* so *pañcama-bhāga* (fifth-part) is also equivalent to $1/5$. Even the word *bhaga* was originally omitted, probably for the sake of metrical convenience, thus only *pañcama* (fifth) could be used for $1/5$. Composite fractions like $2/7$ or $3/8$ were called *dvī-saptama* (two-sevenths) and *tri-astama* (three eighths) respectively.

Fractions were written in the same way as we do now; the numerators above the denominator, but without the line between them. Both the numerator and denominator

were expressed in the decimal place value system. When several fractions occurred in the same problem, they were separated from each other by a vertical and a horizontal line. When a mixed number has to be written the integer was given above the fraction so $2\frac{3}{5}$ was written as $\begin{array}{|c|} \hline 2 \\ \hline 3 \\ \hline 5 \\ \hline \end{array}$.

* REDUCTION TO THE LOWEST TERM AND REDUCTION TO THE COMMON DENOMINATOR:

It was recommended to reduce a fraction to the lowest term before performing operations. The process of reduction was called *aparartana*. This procedure was not included among operations and is not described in mathematical works. Probably it was taught by oral instruction.

The reduction to a common denominator was called *kalā-savarnana*, *savarnana* or *samachheda-vidhi*. This operation was required when the operation addition or subtraction followed. The process was generally mentioned together with these operations. Mahavīra was the first who mentioned the lowest common multiple, he used the term *niruddha* for it. Bhāskara II recommended the process of shortening, but didn't apply the word *niruddha*.

* CLASSES OF FRACTIONS IN COMBINATION

For the sake of shortage of suitable symbolism the expressions with fraction were divided into several classes,

(1) The class bhaga ("simple fractions"), i.e. the form with 2 fractions $(\frac{a}{b} \pm \frac{c}{d})$ with 3

fractions $(\frac{a}{b} \pm \frac{c}{d} \pm \frac{e}{f})$ or with more fractions

was usually written as

$$\begin{array}{|c|c|} \hline a & c \\ \hline b & d \\ \hline \end{array}$$

$$\text{or } \begin{array}{|c|c|} \hline a & \cdot c \\ \hline b & d \\ \hline \end{array}$$

where the dot denotes subtraction. This form with 3 fractions:-

$$\begin{array}{|c|c|c|} \hline a & c & e \\ \hline b & d & f \\ \hline \end{array}$$

$$\text{or } \begin{array}{|c|c|c|} \hline a & \cdot c & \cdot e \\ \hline b & d & f \\ \hline \end{array}$$

(2) The class prabhāga ("fractions of fractions"), i.e. the form $(\frac{a}{b} \cdot \frac{c}{d})$ or $(\frac{a}{b} \cdot \frac{c}{d} \cdot \frac{e}{f})$ which

was written as

$$\begin{array}{|c|c|} \hline a & c \\ \hline b & d \\ \hline \end{array}$$

$$\text{or } \begin{array}{|c|c|c|} \hline a & \cdot c & e \\ \hline b & d & f \\ \hline \end{array}$$

(3) The class bhāganulandha ("fractions in association") included form

a) supa - bhāganulandha (fractions containing associated ~~fractions~~ integers):-

$$(a + \frac{b}{c}) \text{ written as } \begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline \end{array}$$

b) bhāga-bhāganubandhu ("fractions containing associated fractions"), i.e. $(\frac{a}{b} + \frac{c}{d} \cdot \frac{a}{b})$

or $(\frac{a}{b} + \frac{c}{d} \cdot \frac{a}{b} + \frac{e}{f} \cdot (\frac{a}{b} + \frac{c}{d} \cdot \frac{a}{b}))$ in

notation

$$\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline d \\ \hline \end{array}$$

or

$$\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline d \\ \hline e \\ \hline f \\ \hline \end{array}$$

(4) The class bhāgaparāha ("fractions in dissociation") included form

a) sūpa-bhāgaparāha ("fractions containing dissociated integers") meant

$(a - \frac{b}{c})$ written as $\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline \end{array}$

b) bhāga-bhāgaparāha ("fractions containing dissociated fractions"), i.e. $(\frac{a}{b} - \frac{c}{d} \cdot \frac{a}{b})$

or $(\frac{a}{b} - \frac{c}{d} \cdot \frac{a}{b} - \frac{e}{f} \cdot (\frac{a}{b} - \frac{c}{d} \cdot \frac{a}{b}))$ written as

$$\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline d \\ \hline \end{array}$$

or

$$\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline d \\ \hline e \\ \hline f \\ \hline \end{array}$$

(5) The class bhaga-bhaga ("complex fractions") denoted expressions $(a : \frac{b}{c})$ or $(\frac{a}{b} : \frac{c}{d})$

which were written as $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ or $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

There didn't appear any graphic symbol for division, the written form was the same as for bhaganulandha. The fact that division was required followed from the formulation of problems -

(6) Some authors meant extra class bhāga-mātr, i.e. combinations of forms enumerated above. Mahavira remarked that the number of such combinations was 26. As there were five primary classes, he enumerated the total number of combinations.

$$\binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 10 + 10 + 5 + 1 = 26$$

The rules for reduction in the first and the second class are same as rules for addition, subtraction and multiplication, the rule for reduction in fifth class corresponds to the rule for division of fractions. The rule for reduction in class bhaga-bhaganulandha and bhaga-bhagapavaka could be written as:

$$\frac{a}{b} \pm \frac{c}{d} \cdot \frac{a}{b} = \frac{a \cdot (d \pm c)}{b \cdot d} = \frac{a}{b} \cdot \frac{d \pm c}{d}$$

The following example was given by Śaṅkha (c. 870-930) from [Datta, Singh, 1935]. What is the result when half, one-fourth, one-divided by one-third, half plus half of itself and one-third diminished by half of itself are added together? In today's notation it is

$$\frac{1}{2} + \left(\frac{1}{4} \cdot \frac{1}{4}\right) + \left(1 \div \frac{1}{3}\right) + \left(\frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{2} \cdot \frac{1}{3}\right)$$

corresponding old Indian notation was:

1	1	1	1	1	1
2	4	4	1	2	3
			3	1	1
				2	2

The unclarity of notation is obvious

1	1
4	4

could be read as $\left(\frac{1}{4} \cdot \frac{1}{4}\right)$ or as $\left(\frac{1}{4} + \frac{1}{4}\right)$

$\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ could mean $\left(1 \div \frac{1}{3}\right)$ as well as $1 \frac{1}{3}$

The right meaning of notation could be understood only from the formulation of the problem.

INDIA'S

Systematic in "Hellenic" as part of astronomy flourished due to per discoveries → The word Greek & p. words "si" the Lat comes from "touching" of unit from line cu

→ The v and

at d2